

Penetration Test Report – Simple CTF (TryHackMe)

Prepared by: James DeBerry

Date: March 30, 2026

Assessment Type: Black-Box Penetration Test

Executive Summary

A penetration test was conducted against a web application. The assessment identified a critical SQL Injection vulnerability that allowed authentication bypass and system compromise.

The successful attack demonstrates improper input validation and insecure authentication mechanisms.

Scope

- Target IP: 10.892.990.23
- Environment: TryHackMe Lab
- Testing Methodology: External Black-Box Testing

Methodology

The assessment followed a standard penetration testing methodology:

- Reconnaissance & Enumeration
- Vulnerability Identification
- Exploitation
- Post-Exploitation & Impact Analysis

Findings Summary

Severity | Finding

Critical | SQL Injection Authentication Bypass

Detailed Findings

1. SQL Injection Vulnerability

- Severity: Critical
- Port: 80 (HTTP)
- Service: Web Application

Description

The application failed to sanitize user input, allowing SQL Injection to bypass authentication controls.

Evidence

- Login bypass achieved using SQL Injection payload
- Access to admin panel confirmed

Exploitation

Payload used:

' OR 1=1 --

Gained unauthorized access and uploaded web shell.

Proof of Exploitation

System-level access achieved via web shell.

Impact

- Authentication bypass
- Unauthorized access
- Full system compromise

Risk Assessment

- Likelihood: High
- Impact: Critical
- Overall Risk: Critical

Remediation

- Implement input validation
- Use parameterized queries
- Apply secure coding practices

Conclusion

The system is critically vulnerable due to SQL Injection, allowing full compromise.

Appendix

Tools Used

- Nmap
- Burp Suite
- Netcat

Prepared by: James DeBerry
Penetration Tester